RAJALAKSHMI ENGINEERING COLLEGE

Puvithan S 251801252-2025-AIDS

P2

Started on	Friday, 2 October 2026, 10:51 AM
State	Finished
Completed on	Friday, 2 October 2026, 10:52 AM
Time taken	1 min 4 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that A[j] - A[i] = k, i != j.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as 5 - 1 = 4

So Return 1.

For example:

Input	Result
3 1 3 5 4	1

Answer: (penalty regime: 0 %)

```
#include <stdio.h>

int main() {
    int n;
    scanf("%d", &n);
    int arr[n];
    for(int i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }
    int k;
    scanf("%d", &k);

    int i = 0, j = 1;
    while(i < n && j < n) {
        int diff = arr[j] - arr[i];
        if(diff == k && i != j) {
            printf("1\n");  // pair exists
            return 0;
        } else if(diff < k) {
            j++;  // need bigger difference
        } else {
            i++;  // reduce difference
            if(i == j) j++;  // ensure i != j
        }
    }
    printf("0\n");  // no pair found
    return 0;
}

```

	Input	Expected	Got	
✔	3 1 3 5 4	1	1	✔
✔	10 1 4 6 8 12 14 15 20 21 25 1	1	1	✔
✔	10 1 2 3 5 11 14 16 24 28 29 0	0	0	✔
✔	10 0 2 3 7 13 14 15 20 24 25 10	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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P2

Started on	Friday, 2 October 2026, 10:50 AM
State	Finished
Completed on	Friday, 2 October 2026, 10:51 AM
Time taken	1 min 3 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that A[j] - A[i] = k, i != j.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as 5 - 1 = 4

So Return 1.

For example:

Input	Result
3 1 3 5 4	1

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int main() {
    int n;
    scanf("%d", &n);
    int arr[n];
    for(int i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }
    int k;
    scanf("%d", &k);

    int i = 0, j = 1;
    while(i < n && j < n) {
        int diff = arr[j] - arr[i];
        if(diff == k && i != j) {
            printf("1\n");
            return 0;
        } else if(diff < k) {
            j++;
        } else {
            i++;
            if(i == j) j++;  // ensure i != j
        }
    }
    printf("0\n");
    return 0;
}

```

	Input	Expected	Got	
✔	3 1 3 5 4	1	1	✔
✔	10 1 4 6 8 12 14 15 20 21 25 1	1	1	✔
✔	10 1 2 3 5 11 14 16 24 28 29 0	0	0	✔
✔	10 0 2 3 7 13 14 15 20 24 25 10	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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P2

Started on	Friday, 2 October 2026, 10:48 AM
State	Finished
Completed on	Friday, 2 October 2026, 10:49 AM
Time taken	59 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

· The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array
2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

For example:

Input	Result
1 3 10 17 57 6 2 7 10 15 57 246	10 57

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int main() {
    int T;
    scanf("%d", &T);

    while (T--) {
        int N1, N2;
        scanf("%d", &N1);
        int arr1[N1];
        for (int i = 0; i < N1; i++) scanf("%d", &arr1[i]);

        scanf("%d", &N2);
        int arr2[N2];
        for (int i = 0; i < N2; i++) scanf("%d", &arr2[i]);

        int i = 0, j = 0;
        while (i < N1 && j < N2) {
            if (arr1[i] == arr2[j]) {
                printf("%d ", arr1[i]);
                i++; j++;
            } else if (arr1[i] < arr2[j]) {
```

```
            } else if (arr1[i] < arr2[j]) {
                i++;
            } else {
                j++;
            }
        }
        printf("\n");
    }
    return 0;
}
```

	Input	Expected	Got	
✔	1 3 10 17 57 6 2 7 10 15 57 246	10 57	10 57	✔
✔	1 6 1 2 3 4 5 6 2 1 6	1 6	1 6	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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P2

Started on	Friday, 2 October 2026, 10:47 AM
State	Finished
Completed on	Friday, 2 October 2026, 10:48 AM
Time taken	39 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

· The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array
2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

For example:

Input	Result
1 3 10 17 57 6 2 7 10 15 57 246	10 57

Answer: (penalty regime: 0 %)

```
#include <stdio.h>

int main() {
    int T;
    scanf("%d", &T);

    while(T--) {
        int N1, N2;
        scanf("%d", &N1);
        int arr1[N1];
        for(int i=0; i<N1; i++) scanf("%d", &arr1[i]);

        scanf("%d", &N2);
        int arr2[N2];
        for(int i=0; i<N2; i++) scanf("%d", &arr2[i]);

        int i=0, j=0;
        while(i < N1 && j < N2) {
            if(arr1[i] == arr2[j]) {
                printf("%d ", arr1[i]);
                i++; j++;
            } else if(arr1[i] < arr2[j]) {
```

```
            i++;
        } else {
            j++;
        }
    }
    printf("\n");
}
return 0;
}
```

	Input	Expected	Got	
✔	1 3 10 17 57 6 2 7 10 15 57 246	10 57	10 57	✔
✔	1 6 1 2 3 4 5 6 2 1 6	1 6	1 6	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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RAJALAKSHMI ENGINEERING COLLEGE

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P2

Started on	Friday, 2 October 2026, 10:46 AM
State	Finished
Completed on	Friday, 2 October 2026, 10:47 AM
Time taken	33 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

For example:

Input	Result
5 1 1 2 3 4	1

Answer: (penalty regime: 0 %)

```
#include <stdio.h>

int main() {
    int n;
    scanf("%d", &n);

    int arr[n];
    int seen[n+1]; // values are between 1 and n
    for(int i = 0; i <= n; i++) seen[i] = 0;

    for(int i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
        if(seen[arr[i]]) {
            printf("%d\n", arr[i]); // duplicate found
            return 0;
        }
        seen[arr[i]] = 1;
    }

    return 0;
}
```

	Input	Expected	Got	
✔	11 10 9 7 6 5 1 2 3 8 4 7	7	7	✔
✔	5 1 2 3 4 4	4	4	✔
✔	5 1 1 2 3 4	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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RAJALAKSHMI ENGINEERING COLLEGE

Puvithan S 251801252-2025-AIDS

P2

Started on	Friday, 2 October 2026, 10:45 AM
State	Finished
Completed on	Friday, 2 October 2026, 10:46 AM
Time taken	45 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

For example:

Input	Result
5 1 1 2 3 4	1

Answer: (penalty regime: 0 %)

```
#include <stdio.h>

int main() {
    int n;
    scanf("%d", &n);
    int arr[n];
    int seen[n+1]; // values are between 1 and n
    for(int i=0; i<=n; i++) seen[i] = 0;

    for(int i=0; i<n; i++) {
        scanf("%d", &arr[i]);
        if(seen[arr[i]]) {
            printf("%d\n", arr[i]);
            return 0;
        }
        seen[arr[i]] = 1;
    }
    return 0;
}
```

	Input	Expected	Got	
✔	11 10 9 7 6 5 1 2 3 8 4 7	7	7	✔
✔	5 1 2 3 4 4	4	4	✔
✔	5 1 1 2 3 4	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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